

Sarus Operation Planner 1.0.0

The ground control station for ArduPilot aircraft from Sarus Aerospace. It is based on ArduPilot Mission Planner 1.3.83 and keeps all of its screens, features and keyboard shortcuts, with a Sarus look, safer parameter editing and fixes to how it talks to the aircraft.

What's different from Mission Planner

Parameter editing

A value outside ArduPilot's recommended range, such as an unusually high TECS climb rate, is no longer stopped by a question. Sarus writes it as typed, turns the cell amber and puts the recommended range in the tooltip. Parameters that ArduPilot marks read-only can be changed too, once you confirm a warning.

After every write, Sarus asks the aircraft for each value again. Where the aircraft rounded, limited or refused a value, you get a list, and the table shows what the aircraft kept. During testing this caught Q_LOIT_ACC_MAX_M: confirmed at 9.81, actually running at 5.66.

Replies that went missing

Mission Planner could lose an answer from the aircraft when two parts of the program were waiting at the same time, for example a background camera query while you read parameters. One would pick up the other's reply and throw it away, and you would see a timeout even though the aircraft had answered. Sarus registers for each reply before sending the question, so nothing is taken from it. The figures below come from the simulator under heavy load.

OPERATION	BEFORE	SARUS 1.0.0
Parameter reads	6 of 45 lost	0 of 600 lost
Command acknowledgements (arm, mode)	5 of 15 lost	0 of 1,811 lost
Mission upload and download	4 of 10 failed	0 failed

OPERATION	BEFORE	SARUS 1.0.0
Commands with coordinates	2 of 45 lost	0 of 887 lost
Flight-log list during parameter reads	not measured	0 of 326 failed

Smaller fixes

The update check stopped complaining when the server had no update yet, and plugins built by an older version are no longer reused after an upgrade, where they used to fail every second. To learn a release's size, the updater now reads one header instead of downloading the whole file. A legacy rally-point request that got lost is now retried with the right message; before, the retry went out as a fence request. Silent installation with `msiexec /qn` used to hang for ever on the USB driver wizard and now completes. Installing by double-click works exactly as before.

Firmware

Install Firmware lists Sarus firmware builds next to ArduPilot's. They use ArduPilot's board identities, so boards are recognised as usual, and their file names start with "Sarus-". ArduPilot's own firmware is still there, and new ArduPilot releases appear the day they are published.

Look, privacy and coexistence

Sarus Dark is the default theme, with Sarus Light as the alternative; Mission Planner's themes are still in the list. The menu icons and the logo are new. The HUD and the map kept Mission Planner's colours on purpose, because pilots read them at a glance.

Usage statistics are off unless you turn them on. Sarus installs beside Mission Planner rather than over it and keeps its settings, logs and maps in `Documents\Sarus Operation Planner` and `ProgramData\Sarus Operation Planner`. Program updates come from the Sarus releases on GitHub.

How it was tested

The longest test flew 98 automatic QuadPlane VTOL missions in a row over four hours. None failed, every landing was within 0.1 m of home, and memory stayed flat. Every vehicle parameter was written, read back and restored: about 1,180 on

QuadPlane, 1,000 on Copter and 886 on Rover. With the aircraft connected, we opened every SETUP and CONFIG page (46 on QuadPlane, 48 on Copter). Copter flew a mission with take-off, waypoints and return to launch, and Rover drove a square and came back to within 0.2 m of its start.

To see how the program copes with a poor radio, one flight ran with 5% of messages dropped, 300 ms of delay and a 10-second blackout. Telemetry was back 0.3 seconds after the link returned. All main keyboard shortcuts were pressed and checked. Finally, the installer was run silently on fresh Windows Server 2022 and 2025 machines that already had Mission Planner; Sarus started, ran for a minute and uninstalled cleanly, and Mission Planner was unaffected.

All flights were in ArduPilot's simulator. Test with your own aircraft, radios and flight controller on the bench before you fly.

Known limitations

This release runs on Windows only. On screens scaled above 100%, text looks slightly soft, as it does in Mission Planner. Map imagery comes from the same third-party providers Mission Planner uses, under their terms. Like Mission Planner, the installer adds USB drivers and their signing certificate; in a silent installation Windows only adds the drivers whose publisher it already trusts. Until the installer is code-signed, Windows may say "Windows protected your PC". Choose More info, then Run anyway.

VERSION

1.0.0

BASED ON

ArduPilot Mission Planner 1.3.83

TESTED WITH

ArduPilot Plane / QuadPlane (VTOL), Copter and Rover 4.7, in the simulator

WINDOWS

10 or 11, 64-bit

NEEDS

.NET Framework 4.7.2 or later, about 500 MB disk, 4 GB RAM

LICENCE

GNU GPL v3, no warranty

SOURCE CODE

github.com/LikhitS/SarusOperationPlanner

PUBLISHER

Sarus Aerospace Pvt Ltd · sarusaerospace.in

Sarus Operation Planner is based on Mission Planner by Michael Osborne and the ArduPilot contributors. ArduPilot is a trademark of its owners and does not endorse this product.